Exercise 1 :

Calculate $P(A)$, $P(B)$, $P(A|B)$ from the joint probability distribution $P(A, B)$ given in Table 1.

	b_1	b_2	b_3
a_1	0.05	0.10	0.05
a_2	0.15	0.00	0.25
a_3	0.10	0.20	0.10

Table 1: The joint probability distribution for $P(A, B)$.

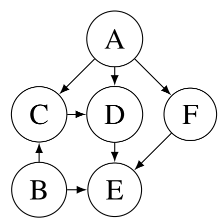


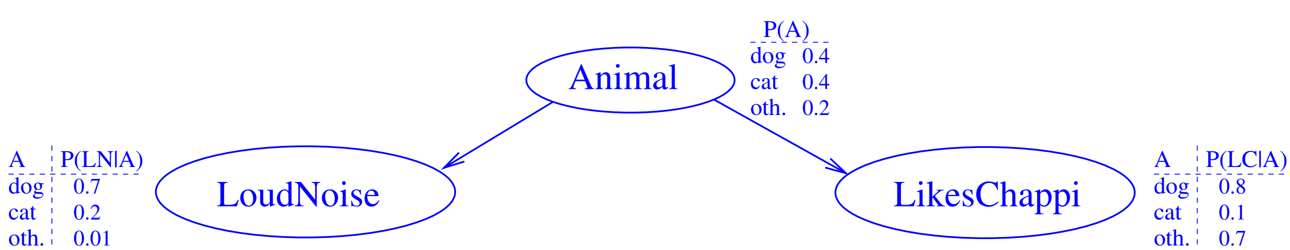
Figure 1: A Bayesian network for Exercise 2.

Exercise 2 :

Consider the network in Figure 1. Which conditional probability tables must be specified to turn the graph into a Bayesian network? Assuming that all variables are binary, what is the size of those tables (i.e., how many values do we need to specify?)

Exercise 3 :

Consider the following Bayesian network BN:



Use this network to compute the following probabilities:

(a) $P(loudnoise, dog, likeschappi)$.

(b) $P(loudnoise, other, likeschappi)$.

(c) $P(\neg loudnoise, dog, likeschappi)$.

Include intermediate steps at a level of granularity as in the lecture slides examples. In particular, for each of the probabilities P demanded in (a) – (c), write down *which* probabilities provided in BN can be combined *how* to obtain P.

Exercise 4 :

We want to construct a Bayesian network for the following random variables (all with domain *true*,*false*):

Mexico	Person X has recently travelled to Mexico
Svine_flu_infection	Person X has been infected with the svine flu virus
Vaccination	Person X has been vaccinated against svine flu
Svine_flu_sick	Person X is sick from svine flu
Fever	Person X has fever

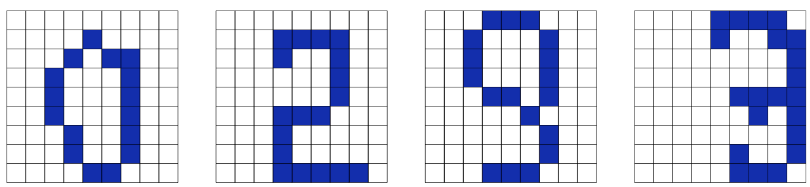
a. Use the above ordering of the variables to determine a Baysian network structure based on the chain rule and conditional independence relations.

b. Repeat the construction with the alternative variable ordering:

Vaccination,*Mexico*,*Svine_flu_sick*,*Fever*,*Svine_flu_infection*

Exercise 5 :

Design a Bayesian network that can be used to recognize handwritten digits 0,1,2,. . . ,9 from scanned, pixelated images like these:



- What are hypothesis and information variables?
- Could there be any useful mediating variables (consider e.g. the last image above)?
- How could you design a network structure
  - so that the conditional independencies are (approximately) reasonable
  - so that specification and inference complexity remain feasible
- How do you fill in the conditional probability tables?

Exercise 6 :

For 10000 emails in your inbox you determine the values of the following three boolean variables:

Spam	the email is spam
Caps	the subject line is in all capital letters
Pills	Body of the message contains the word “pills”

You obtain the following counts:

	Caps			
	yes		*no*	
	Pills		*Pills*	
Spam	*yes*	*no*	*yes*	*no*
yes	150	850	600	3400
no	1	99	49	4851

Are

- *Spam* and *Caps* independent?
- *Pills* and *Caps* independent?
- *Pills* and *Caps* independent given *Spam*?
- *Spam* and *Caps* independent given *Pills*?

Exercise 7 :

Consider the experiment of flipping a coin, and if it lands heads, rolling a four-sided die, and if it lands tails, rolling a six-sided die.

(i) Suppose that the coin and both dice are fair. As we are interested only in the number rolled by the die, and the possible worlds $\mathcal{S}_A$ for the experiment could thus be the numbers from 1 to 6. Another set of possible worlds could be $\mathcal{S}_B = \{t1, \ldots, t6, h1, \ldots, h4\}$, with for example $t2$ meaning "tails and a roll of 2" and $h4$ meaning "heads and a roll of 4." Choose either $\mathcal{S}_A$ or $\mathcal{S}_B$ and associate probabilities with it. According to your chosen set of possible worlds and probability distribution, what is the probability of rolling either 3 or 5.

(ii) Let $\mathcal{S}_B$ be defined as above, but with a loaded coin and loaded dice. A probability distribution is given in Table 2. What is the probability that the loaded coin lands "tails"? What is the conditional probability of rolling a 4, given that the coin lands tails? Which of the loaded dice has the highest chance of rolling 4 or more?

$t1$	$\frac{5}{18}$	$t6$	$\frac{1}{18}$
$t2$	$\frac{1}{9}$	$h1$	$\frac{1}{24}$
$t3$	$\frac{1}{9}$	$h2$	$\frac{1}{24}$
$t4$	$\frac{1}{18}$	$h3$	$\frac{1}{8}$
$t5$	$\frac{1}{18}$	$h4$	$\frac{1}{8}$

Table 2: Probabilities for $\mathcal{S}_B$.

Exercise 8 :

Consider the binary variable A and the ternary variable B. Assume that B has the probability distribution $P(B) = (0.1, 0.5, 0.4)$ and that A has the conditional probability distribution given in Table 3.

Questions:

(i) Verify that Table 3 specifies a valid conditional probability distribution.

	b_1	b_2	b_3
a_1	0.1	0.7	0.6
a_2	0.9	0.3	0.4

Table 3: The conditional probability distribution $P(A|B)$.

(ii) Calculate $P(B|A)$. *Hint:* Consider Bayes rule illustrated by the temperature-sensor example that we discussed in the lecture

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Exercise 9 :

A routine DNA test is performed on a person (this exercise is set in the not too distant future!). The test T gives a positive result for a rare genetic mutation M linked to Alzheimer's disease. The mutation is present in only 1 in a million people. The test is 99.99% accurate, i.e. it will give a wrong result in 1 out of 10000 tests performed. Should the person be worried, i.e., what is the probability that the person has the mutation given that the test showed a positive result?

Hint: This is partly a modeling exercises and partly a calculation exercise. First you need to formalize the problem:

- What are the relevant variables and what states do they have?
- Based on the description above, what probability distributions can you infer for the variables?

Based on this formalization, you need to find the rules required to answer the question about the probability of a mutation given a positive test result.

Exercise 10 :

Assume it is your responsibility to monitor volcanic eruptions. You receive data from two different stations (seismometers), S_1 and S_2. Each S_i is modeled as a Boolean variable where "true" stands for "I detected an eruption" and "false" stands for "I did not detect an eruption". The seismometers are not fully reliable, however; they may not detect an eruption even though there was one, and they may mistake an earthquake for an eruption of a volcano. We model this situation with two additional Boolean variables: V for volcanic eruption, and E for Earthquake.

Use the algorithm from the lecture to construct a Bayesian network for these 4 variables. Do so for the following two variable orders:

(a) $X_1 = V, X_2 = E, X_3 = S_1, X_4 = S_2$.

(b) $X_1 = S_1, X_2 = S_2, X_3 = E, X_4 = V$.

For each of these orders, draw the resulting Bayesian network. Justify your design, i.e., for each variable X_i added to the network explain why the set of parents you give X_i are needed, and why they are sufficient.

Exercise 11 :

Table 4 describes a test T for an event A. The number 0.01 is the frequency of *false negatives*, and the number 0.001 is the frequency of *false positives*.

	$A = yes$	$A = no$
$T = yes$	0.99	0.001
$T = no$	0.01	0.999

Table 4: Table for Exercise 11. Conditional probabilities $P(T \mid A)$ characterizing test T for A.

	b_1	b_2
a_1	$(0.006, 0.054)$	$(0.048, 0.432)$
a_2	$(0.014, 0.126)$	$(0.032, 0.288)$

Table 5: $P(A, B, C)$ for Exercise 12.

(*i*) The police can order a blood test on drivers under the suspicion of having consumed too much alcohol. The test has the above characteristics. Experience says that 20% of the drivers under suspicion do in fact drive with too much alcohol in their blood. A suspicious driver has a positive blood test. What is the probability that the driver is guilty of driving under the influence of alcohol?

(*ii*) The police block a road, take blood samples of all drivers, and use the same test. It is estimated that one out of 1,000 drivers have too much alcohol in their blood. A driver has a positive test result. What is the probability that the driver is guilty of driving under the influence of alcohol?

Exercise 12 :

In Table 5, a joint probability table for the binary variables A, B, and C is given.

- Calculate $P(B, C)$ and $P(B)$.
- Are A and C independent given B?
